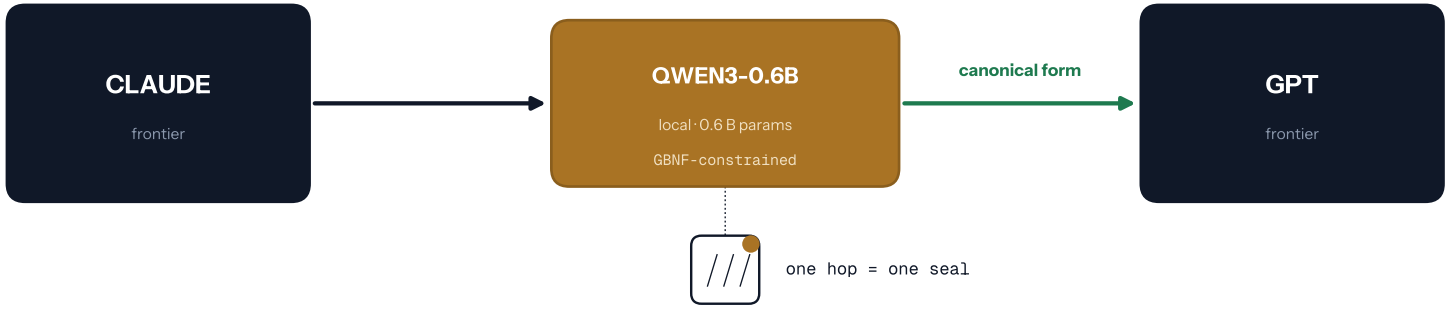
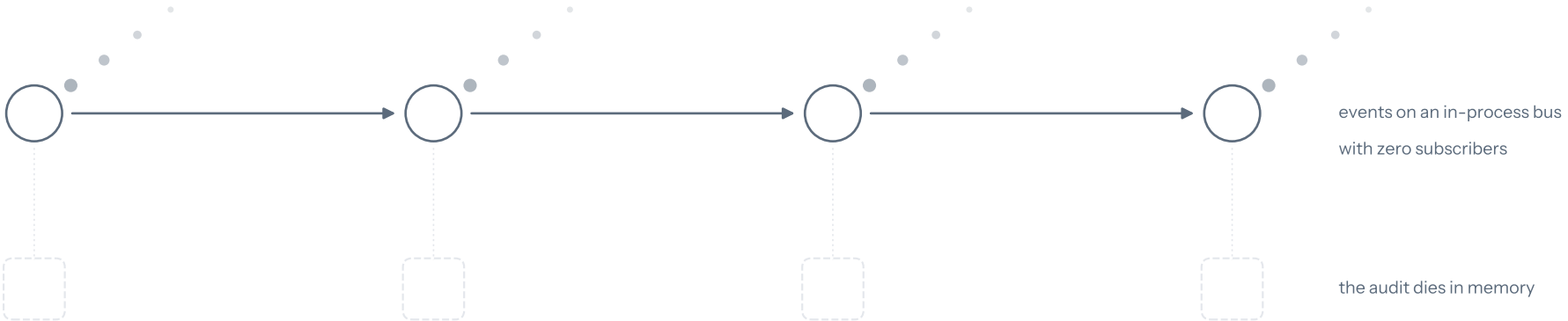
Small models that carry information between the big ones — and prove it.

Every hop is stamped, sealed, re-verifiable offline.



Fil B — shipped: B.0 → B.4 · 6 commits · 04759dd → 02eccdb

Before: the hop happened. Nothing proved it.



The claim ‹relayed by the small model›

unverifiable — payload truncated at 120 chars

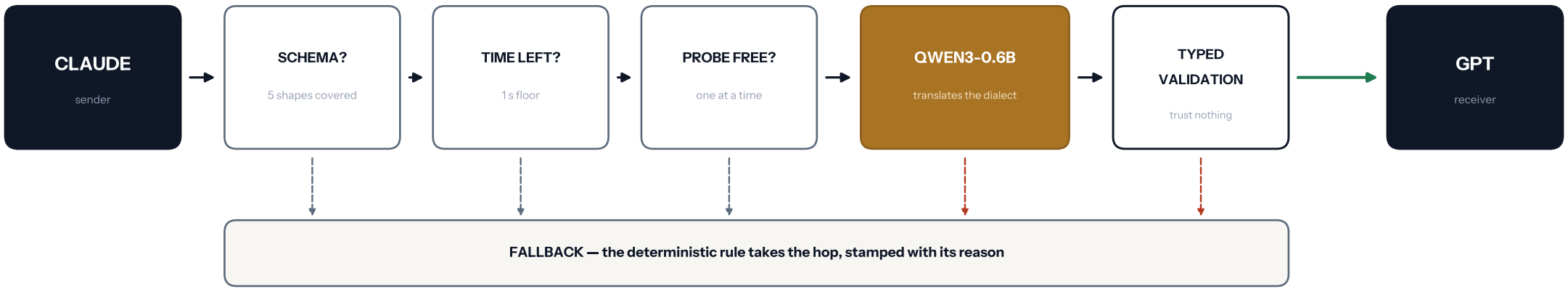
audit_chain_hash

= the mission id, reformatted. Not a hash.

A hung translator

could steal 30 s from a 2 s mission

Three guards, one small model, one stamp.



SLM
the model translated — validated

RULE
no model wired

SLM_DEGRADED_TO_RULE
replaced — reason sealed

One env var wires it. Without it — the default path is byte-identical.

Forge one stamp — the re-proof breaks. Localized.

each hop committed one by one into the mission's HMAC chain



HMAC-sealed

per-hop node in the graph — engine, reason, input/output fingerprints (blake3), tokens, latency

Re-verifiable

offline, with the tenant key — no need to trust the server that produced it

slm_mediated

true if and only if EVERY hop is SLM — one degradation and the claim drops, visibly

Honest by construction: a degraded hop can never be presented as a model hop.

The numbers.

1 016

tests green

45 suites · 0 failures

43 → 19

adversarial review

raw findings → real defects fixed

45 / 0

completeness audit

checks run / gaps found

6

commits

B.0 → B.4 + audit

4 adversarial review passes — agents paid to refute my own work; one caught a defect that would have produced a FALSE TAMPER signal on an honest run. 6 independent verifiers audited B.0→B.4 against the spec: 45 checks, zero gaps. Every number above is reproducible from the repo.

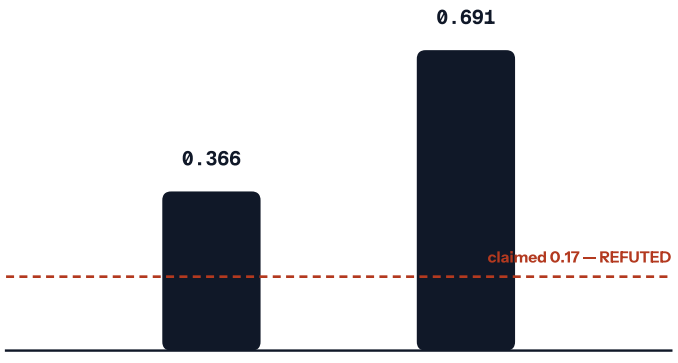
Not a demo. Served, opt-in, with the default path proven byte-identical.

Three of my own hypotheses died on measurement.

I shipped the negative results with the same rigor as the positive ones.

Lipschitz bound — meta-swarm

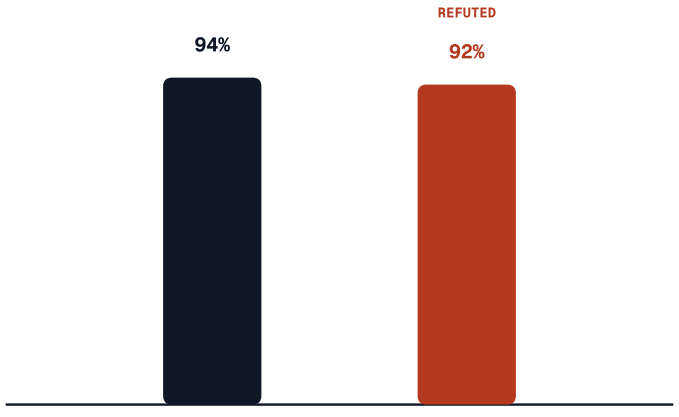
measured constant (lower = tighter)



Our own doc claimed an analytic bound of 0.17. Both measurements are far above it. The certificate is refuted — the honest bound is the measured one.

HB.3 — arithmetic re-execution verifier

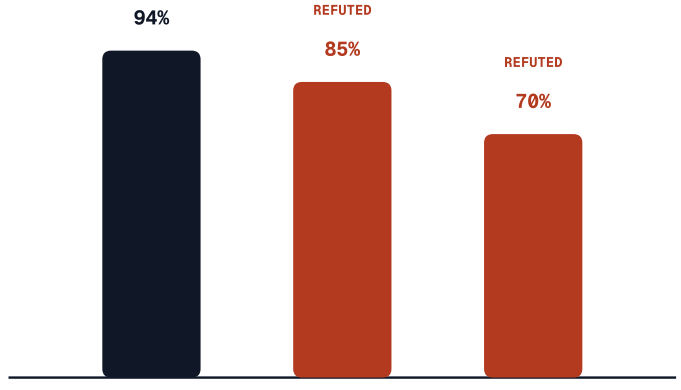
% correct @ n=50



The deployable verifier lands BELOW the baseline it was meant to beat. The residual gap is modeling, not calculation.

HB.4 — robust aggregation

% correct, phi4 x math

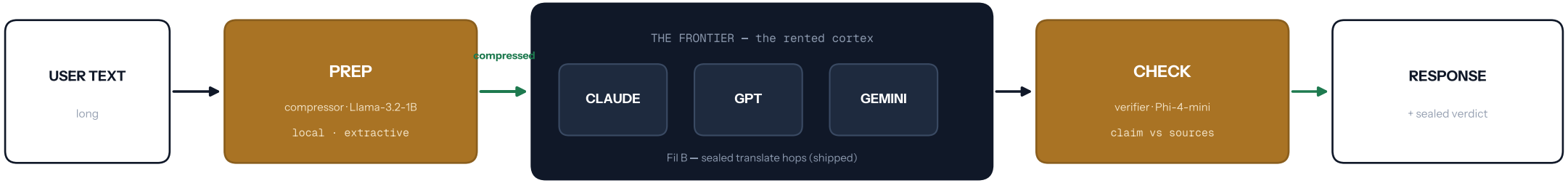


Errors are systematically skewed (skew +8.72), not symmetric. No deployable aggregation closes the gap. Two hypotheses dead, one arc closed.

What you get: systems where the claim is only as strong as the receipt behind it.

C.1 — the sovereign sandwich.

ARCHITECTURE DONE — NOT CODED



same receipt:

prep_0 · reform_* · check_0

Local intelligence frames every rented call

compression before, verification after — both sealed

The cost is in the same receipt

the reader computes gains minus cost themselves

Gains will be MEASURED before they are claimed

no ×8 marketing number — heuristic estimates, labeled

The claim is only as strong as

the receipt behind it.



one hop = one seal = one verdict

Fil B shipped · 1 016 tests · 0 gaps · 3 hypotheses refuted and published